VIT-AP UNIVERSITY	Final Assessment Test – Winter (2024-25) - April 2025	
	Maximum Marks: 100	Duration: 3 Hours
Course Code: CSE1008	Course Title: Theory of Computation	
Set No: 1	Exam Type : **Closed Book**	School: SCOPE
Date: 08/05/2025	Slot: D2	Session: FN

Keeping mobile phone/smart watch, even in 'off' position is treated as exam malpractice

General Instructions if any:

1. **"*fx* series" - non Programmable calculator are permitted : NO**
2. **Reference tables permitted : NO**

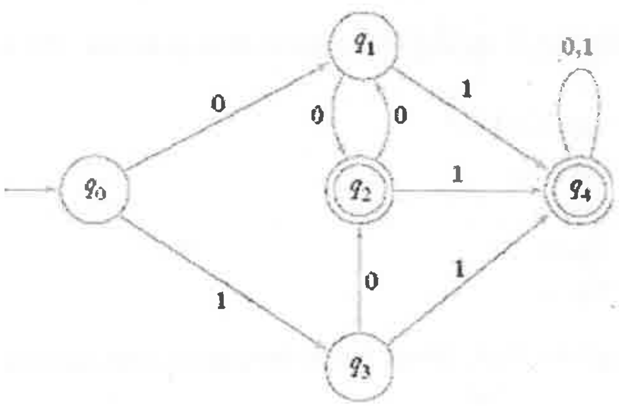
Answer any <u>TEN</u> Questions, Each Question Carries 10 Marks (10×10=100 Marks)

1. Consider the following ϵ-NFA

δ	ϵ	a	b
→p	{r}	{q}	{p,r}
q	Φ	{p}	Φ
*r	{p,q}	{r}	{p}

Compute the ϵ-closure of each state. Convert the given ϵ-NFA to an NFA. Write the steps to convert ϵ-NFA to an NFA neatly and determine the language accepted by NFA. **(10 M)**

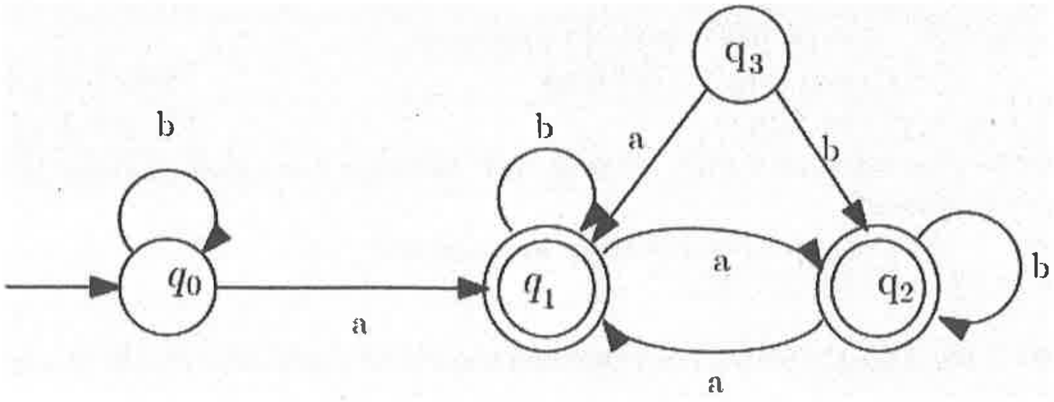
2. Minimize the following DFA by merging equivalent states. Construct the minimized DFA with its new set of states. Describe the tuple form and transitions. **(10 M)**



3. Design a pushdown automata (PDA) to accept the language $L = \{a^m b^n c^{m+n} \mid n,m \geq 0\}$. Describe the steps clearly for design and draw transition diagram of the PDA. Show the instantaneous descriptions (IDs) for the string w = "aaabbbcccccc" of the pushdown automata. **(10 M)**

4. Construct a non-deterministic finite automata (NFA) for the regular expression (11)* + (111)* + 101 on $\Sigma = \{0,1\}$. **(10 M)**

5. Using Arden's theorem, determine the regular expression for the DFA shown below. Explain the step by step procedure for converting the finite automata to regular expression. **(10 M)**



6. The following grammar generates prefix expressions with operands x and y and binary operators +, -, and *: **(10 M)**

$$E \rightarrow +EE \mid *EE \mid -EE \mid x \mid y$$

Find leftmost and rightmost derivations, and a derivation tree for the string +*-xyxy.

7. Consider the context-free grammar G defined by the productions: **(10 M)**

$$S \rightarrow bS \mid aA \mid \epsilon$$
$$A \rightarrow aS \mid bA$$

Determine the language generated by G given above. Check whether G is ambiguous or not for the string 'abbab' and 'baaba'.

8. Convert the PDA $P = (\{q_0, q_1\}, \{a,b\}, \{X, Z_0\}, \delta, q_0, Z_0)$ to a context-free grammar, if δ is given by: **(10 M)**

$$\delta(q_0, a, Z_0) = (q_0, XZ_0)$$
$$\delta(q_0, a, X) = (q_0, XX)$$
$$\delta(q_0, b, X) = (q_1, \epsilon)$$
$$\delta(q_1, b, X) = (q_1, \epsilon)$$
$$\delta(q_1, \epsilon, Z_0) = (q_1, \epsilon)$$

Also, determine the language accepted by given PDA. Show the instantaneous descriptions (IDs) for the string w = "aaabbb" of the PDA.

9. Determine whether the given instances of post correspondence problem (PCP) has a solution: **(10 M)**

a) A = {abb, aa, aaa} and B = {bba, aaa, aa}.
b) A = {10, 011, 101} and B = {101, 11, 011}.

If yes, then give the step by step procedure to determine the finite sequence.

10. Convert the following context free grammars into Greibach Normal Form (GNF). **(10 M)**

$$S \rightarrow CA \mid BB$$
$$B \rightarrow b \mid SB$$
$$C \rightarrow b$$
$$A \rightarrow a$$

11. Show that the language $L = \{ww^r w \mid w \in \{a,b\}^*\}$ is not a context free language using pumping lemma. **(10 M)**

12. Design a turing machine (TM) that accepts the language given below: **(10 M)**

$$L = \{ww^r \mid w \in \{a,b\}^*\}$$

Show the steps and operations on the tape. Show the instantaneous descriptions (IDs) for the string w = "abbbba" of the turing machine.

QP MAPPING

Q. No.	E/A/T	Module Number	Marks	BL	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped
Q1	E	1	10	3	CO1	1, 3	1, 4	1, 2, 3
Q2	E	1	10	2	CO1	1, 3	1, 4	1, 2, 3
Q3	A	4	10	1	CO3	3, 4, 5	1, 2, 3, 4	1, 2, 3
Q4	T	2	10	2	CO2	1, 2, 3	1, 4	1, 2, 3
Q5	T	2	10	1	CO2	1, 2, 3	1, 4	1, 2, 3
Q6	A	3	10	6	CO2	1, 2, 3	1, 4	1, 2, 3
Q7	A	3	10	3	CO2	1, 2, 3	1, 4	1, 2, 3
Q8	A	4	10	6	CO3	3, 4, 5	1, 2, 3, 4	1, 2, 3
Q9	E	6	10	5	CO4	3, 4, 5	1, 2, 3, 4	1, 2, 3
Q10	T	5	10	4	CO3	3, 4, 5	1, 2, 3, 4	1, 2, 3
Q11	A	5	10	4	CO3	3, 4, 5	1, 2, 3, 4	1, 2, 3
Q12	T	6	10	5	CO4	3, 4, 5	1, 2, 3, 4	1, 2, 3

VIT-AP UNIVERSITY	**Final Assessment Test – Winter (2024-25) - April 2025**	
	Maximum Marks: 100	Duration: 3 Hours
Course Code: CSE1008	Course Title: Theory of Computation	
Set No: 04	Exam Type: **Closed Book**	School: SCOPE
Date: 29/04/2025	Slot: A1	Session: FN

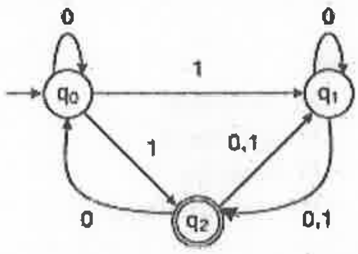
Keeping mobile phone/smart watch, even in 'off' position is treated as exam malpractice

General Instructions if any:

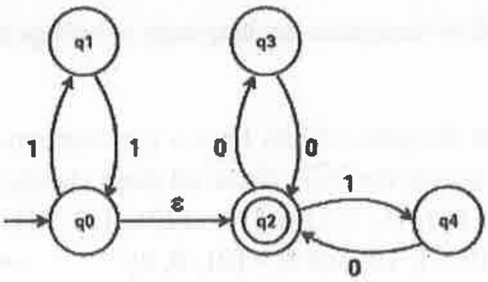
1. **"*fx* series" - non Programmable calculator are permitted : NO**
2. **Reference tables permitted : NO**

PART – A: Answer any TEN Questions, Each Question Carries 10 Marks (10×10=100 Marks)

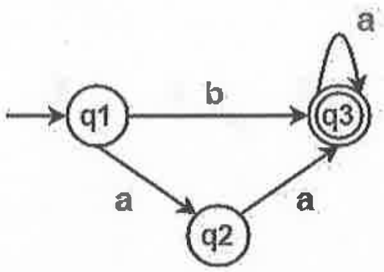
1. Convert the given NFA to a DFA. Write all the steps for the conversion. Check whether the string 10001 is accepted by the DFA or not. **[10 M]**



2. Convert the given ε-NFA to a DFA. **[10 M]**



3. Apply Arden's Theorem to convert the given finite automata to a Regular Expression. **[10 M]**



4. Consider the language L = $\{a^p \mid \text{p is a prime number}\}$, which contains strings like aa, aaa, aaaaa, etc. Show that it is not regular. **[10 M]**

5. Design the CFG for a basic mathematical calculator which can perform addition, subtraction, multiplication, and division. Derive the valid strings (1+2)+6 for addition and 2*(4*5) multiplication. **[10 M]**

6. Derive the different derivation trees for the given production rules of a grammar for the string 'ppqq'. **[10 M]**

$$S \rightarrow A \mid B$$
$$A \rightarrow pAq \mid AA \mid \varepsilon$$
$$B \rightarrow pBq \mid BB \mid pq$$

Consider the string "ppqq" to check whether the above grammar is ambiguous or not.

7. Construct a PDA for the language where the number of a's are equal to the number of c's, while b can appear any number of times between them. Show all the tuples of the PDA. Check whether the string aabbbcc is accepted by the PDA or not. **[10 M]**

8. Construct a PDA for the language $L=\{ww^R\}$, where $\Sigma = \{0,1\}$. Check the acceptance of the strings 101101 and 10101. **[10 M]**

9. Convert the given CFG into CNF. **[10 M]**

$$M \rightarrow NNOP$$
$$N \rightarrow mNn \mid \varepsilon$$
$$O \rightarrow mo \mid m$$
$$P \rightarrow mPm \mid nPn \mid \varepsilon$$

10. Convert the given CFG into GNF. **[10 M]**

$$S \rightarrow A\,a \mid a$$
$$A \rightarrow S\,b \mid b$$

11. Design a TM to recognize the language of strings of the form V^nT^n where $n \geq 1$. Provide ID for V^3T^3. **[10 M]**

12. Find whether the pairs of lists have a post correspondence solution? Find a sequence of indices (if any) that solves the PCP. Show all steps clearly. **[10 M]**
    a. M = (122, 11, 111) and N = (221, 111, 11)
    b. A = [0, 01, 10] and B = [01, 0, 0]

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VIT-AP UNIVERSITY	Final Assessment Test – Winter (2024-25) - April 2025	
	Maximum Marks: 100	Duration: 3 Hours
Course Code: CSE1008	Course Title: Theory of Computation	
Set No: 7	Exam Type : **Closed Book**	School: SCOPE
Date: 02/05/2025	Slot: B1	Session: FN

Keeping mobile phone/smart watch, even in 'off' position is treated as exam malpractice

General Instructions if any:

1. **"*fx* series" - non Programmable calculator are permitted : NO**
2. **Reference tables permitted : NO (if Yes, Please specify:)**

PART – A: Answer any TEN Questions, Each Question Carries 10 Marks (10×10=100 Marks)

1. Convert the following NFA to a DFA and construct DFA states, tuple form and transition diagram for a DFA. **(10 M)**

	0	1
→a	{a,b}	{a}
b	{c,d}	{e}
c	{a,c}	{e}
*d	-	-
*e	-	-

2. Minimize the following DFA and draw the transition diagram of minimized DFA. **(10 M)**

	0	1
→A	B	A
B	A	C
C	D	B
D	D	A
*E	D	F
F	G	E
G	F	G
H	G	D

3. Convert the following DFA into regular expression using Arden's theorem method. **(10 M)**

δ	0	1
$\rightarrow q_1$	q_2	q_3
q_2	q_1	q_3
$*q_3$	q_2	q_1

4. Prove that the following are not regular languages. **(10 M)**

a) $\{0^n \mid n \text{ is a perfect cube}\}$
b) $\{0^n \mid n \text{ is a power of } 2\}$

5. $L_1=\{a^nb\}$ and $L_2=\{ab^m\}$ where $n,m\geq 0$; Write the language $L_1 \cap L_2$ and perform closure under intersection. Draw the corresponding finite automata. **(10 M)**

6. Consider the following grammar **(10 M)**

$S \rightarrow A1B$

$A \rightarrow 0A \mid \varepsilon$

$B \rightarrow 0B \mid 1B \mid \varepsilon$

a) Show that this grammar is unambiguous.

b) Convert the CFG to PDA accepting by empty stack.

7. Design a PDA accepting by final stack for the following language $L=\{a^ib^jc^k, i,j,k>0 \text{ and } i=k\}$ and perform ID for the string w=aaabbccc. **(10 M)**

8. Let PDA $P_N=(\{q0,q1\}, \{0,1\},\{X,Z0\}, \delta, q0 , Z0, \delta)$ where δ is given by

$\delta(q0,1, Z0) =\{(q0, XZ0)\}$

$\delta(q0,1, X) =\{(q0, XX)\}$

$\delta(q0,0, X) =\{(q1, X)\}$

$\delta(q0, \varepsilon, X) =\{(q0, \varepsilon)\}$

$\delta(q1, 1, X) =\{(q1, \varepsilon)\}$

$\delta(q1, 0, Z0) -\{(q0, Z0)\}$

Convert the PDA accepting by empty stack to accepting by final state. **(10 M)**

9. Convert the CFG to Chomsky Normal Form. **(10 M)**

$S\rightarrow aAa \mid bBb \mid \varepsilon$

$A\rightarrow C \mid a$

$B\rightarrow C \mid b$

$C\rightarrow CDE \mid \varepsilon$

$D\rightarrow A \mid B \mid ab$

10. Consider the following grammar, **(10 M)**

$A \rightarrow BA \mid a$

$B \rightarrow AB \mid b$

Convert this grammar into the form $A\rightarrow a\alpha$, where 'a' is a terminal and α is a strings of zero or more variables.

11. Let convert the TM into an instance of MPCP using the input string w=01. **(10 M)**

M={{q1,q2,q3},{0,1},{0,1,B}, δ,B,{q3}}

where δ:

δ	0	1	B
→q1	(q2,1,R)	(q2,0,L)	(q2,1,L)
q2	(q3,0,L)	(q1,0,R)	(q2,0,R)
*q3	-	-	-

12. Design a Turing machine for palindrome over {0,1} for even and odd string length. **(10 M)**
Perform ID for **"10101"** and **"1111"**.

QP MAPPING

Q. No.	E/A/T	Module Number	Marks	BL	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped
Q1	E	1	10	2	CO1	PO1, PO3		
Q2	T	1	10	3	CO1	PO1, PO3		
Q3	A	2	10	2	CO2	PO1, PO2, PO3		
Q4	E	2	10	2	CO2	PO1, PO2, PO3		
Q5	A	3	10	3	CO2	PO1, PO2, PO3		
Q6	T	3	10	3	CO2	PO1, PO2, PO3		
Q7	A	4	10	3	CO3	PO3, PO4, PO5		
Q8	A	4	10	3	CO3	PO3, PO4, PO5		
Q9	A	5	10	3	CO3	PO3, PO4, PO5,		
Q10	A	5	10	3	CO3	PO3, PO4, PO5		
Q11	A	6	10	3	CO4	PO3, PO4, PO5		
Q12	A	6	10	3	CO4	PO3, PO4, PO5		

VIT-AP UNIVERSITY	Final Assessment Test – Winter (2024-25) - April / May 2025	
	Maximum Marks: 100	Duration: 3 Hours
Course Code: CSE1008	Course Title: Theory of Computation	
Set No: 08	Exam Type : **Closed Book**	School: SCOPE
Date: 07/05/2025	Slot: D1	Session: FN

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General Instructions if any:

1. **"*fx* series" - non Programmable calculator are permitted : NO**
2. **Reference tables permitted : NO**

PART – A: Answer any <u>TEN</u> Questions, Each Question Carries 10 Marks (10×10=100 Marks)

1. Design a Deterministic Finite Automaton (DFA) to accept the language L over the alphabet $\Sigma = \{a, b\}$, where each string in L contains: an even number of a's, and an odd number of b's. Provide the formal definition of the DFA and draw its state transition diagram. **(10 M)**

2. Minimize the following DFA by merging equivalent states. Present the minimized DFA with its new set of states and transitions. **(10 M)**

Current State	Input Symbol	
	a	b
→ q_0	q_1	q_2
q_1	q_4	q_3
q_2	q_4	q_3
*q_3	q_5	q_6
*q_4	q_7	q_6
q_5	q_3	q_6
q_6	q_6	q_6
q_7	q_4	q_6

3. Construct NFA for the regular expression 01+101(11*0)*0. Explain the step-by-step procedure. **(10 M)**

4. Let L= $\{a^{2^n} \mid n \geq 0\}$, i.e., the set of strings consisting of a's whose lengths are powers of 2. Prove that L is not regular using the Pumping Lemma. **(10 M)**

5. Design a Context-Free Grammar (CFG) for the language L over the alphabet $\Sigma = \{a, b\}$, where every string in L contains twice the number of a's as b's. Choose a valid example string from the language and show the leftmost derivation and rightmost derivation. Demonstrate that the CFG is ambiguous. **(10 M)**

6. Consider the following Context-Free Grammar (CFG):
   $S \rightarrow (S) \mid S \supset S \mid \sim S \mid p \mid q$
   Given the string **w = (~ ~ p ⊃ (p ⊃ ~ ~ q))**, perform the following:
   (a) Find the leftmost derivation of w.
   (b) Find the rightmost derivation of w.
   (c) Construct the derivation (parse) tree for w. **(10 M)**

7. Design a PDA for the Language $L = \{a^n ccb^n \mid n \geq 0\}$ **(10 M)**

8. Given the PDA $M = (\{q_1, q_2\}, \{a, b\}, \{Z, P, Q\}, \delta, q_1, Z, \varphi)$, convert into CFG. **(10 M)**

$\delta(q_1, a, Z) = (q_1, PZ)$
$\delta(q_1, a, P) = (q_1, PP)$
$\delta(q_1, a, Q) = (q_1, \varepsilon)$
$\delta(q_1, b, Z) = (q_1, QZ)$
$\delta(q_1, b, Q) = (q_1, QQ)$
$\delta(q_1, b, P) = (q_1, \varepsilon)$
$\delta(q_1, \varepsilon, P) = (q_2, \varepsilon)$
$\delta(q_2, \varepsilon, Z) = (q_2, \varepsilon)$

9. Let $L=\{a^{n^2} \mid n \geq 0\}$, i.e., the set of strings consisting of a's, where the length of each string is a perfect square. Prove that the language L is not context-free using the Pumping Lemma for Context-Free Languages (CFLs). **(10 M)**

10. Given the grammar

$S \rightarrow ASB \mid \varepsilon$
$A \rightarrow aAS \mid a \mid \varepsilon$
$B \rightarrow SbS \mid A \mid bb$

(a) Convert the grammar into Chomsky Normal Form (CNF).
(b) Convert the grammar into Greibach Normal Form (GNF). **(10 M)**

11. Design a Turing Machine that computes the 2's complement of a binary number. The binary number is given as input on the tape (most significant bit on the left), and the output should replace the input with its 2's complement. Write the sequence of Instantaneous Descriptions (IDs) for the Turing Machine when it processes the input 1100100. **(10 M)**

12. Determine whether there exists a solution to the Post Correspondence Problem (PCP) for the given lists A and B. Each list contains strings over the binary alphabet {0, 1}.

i. List A = (1, 10111, 10) and List B = (111, 10, 0)
ii. List A = (100, 0, 1) and List B = (1, 100, 00) **(10 M)**